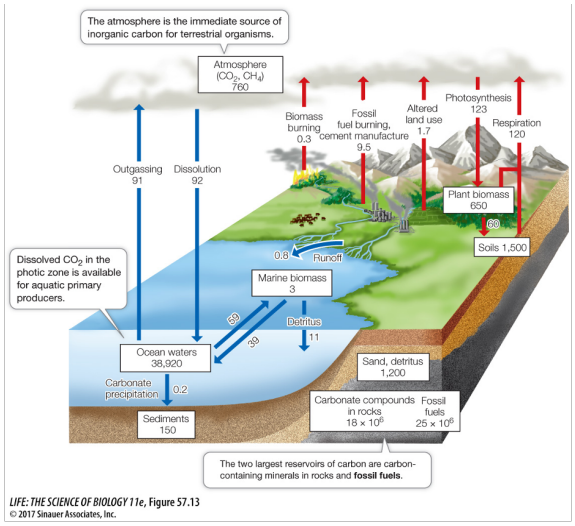
Study Guide for Ecosystems

This study guide provides examples for the topics that you might see on the final exam over the Ecosystems lectures. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and your homework questions to review all the topics that have been covered.

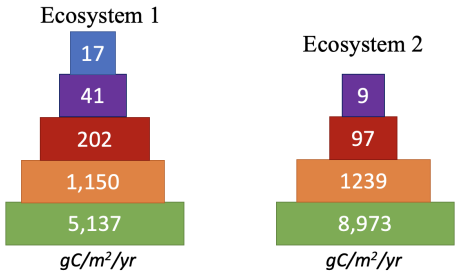
1. From the carbon cycle diagram below, which of the following is/are a process by which carbon can move out of the atmospheric pool? (Select all that apply)
    a. biomass burning
    b. fossil fuel burning
    c. altered land use
    d. photosynthesis
    e. respiration
    f. outgassing
    g. dissolution
    h. runoff



2. Fill in the blanks: On land, net primary productivity is primarily driven by [blank1], while in the ocean, net primary productivity is primarily determined by [blank2].

    Answer options: the concentration of greenhouse gasses in the atmosphere, the amount of solar radiation, the frequency of disturbance events, the amount of available nutrients, the species diversity of the plant community, average annual precipitation

3. You are studying the energy available at different trophic levels in two ecosystems and you develop the energy pyramids below to represent the energy availability at each trophic level in each ecosystem (in units of grams of carbon available per square-meter of habitat per year). Why is Ecosystem 1 capable of supporting five trophic levels, when Ecosystem 2 can only support four?
    a. Because Ecosystem 1 has higher efficiency of energy transfer across trophic levels than Ecosystem 2
    b. Because Ecosystem 1 has more energy entering the ecosystem through the producer trophic level than Ecosystem 2
    c. Because Ecosystem 1 follows Lindeman's Law and Ecosystem 2 does not
    d. Because Ecosystem 1 is a terrestrial ecosystem and Ecosystem 2 is an aquatic ecosystem



4. In class, we discussed the conditional interaction between plants and mycorrhizal fungi. What conditions determine whether this interaction is a mutualism or parasitism of the plant by the fungi?
    a. whether the plants are growing in high or low elevation ecosystems
    b. whether the two species occur sympatrically or allopatrically
    c. whether the ecosystem is experiencing primary or secondary succession
    d. whether the ecosystem is experiencing a top down or bottom up cascade
    e. whether the soil nutrient availability is high or low

5. Some butterflies illustrate a form of mimicry, in which non-toxic species mimic the coloration patterns of toxic butterflies. In this type of mimicry, the non-toxic species benefits because predators assume it is as harmful to consume as the toxic species it mimics. When considering the ecological efficiency between the trophic levels of the non-toxic mimic and its predators, this form of mimicry most directly affects which energy efficiency step?
    a. Consumption efficiency
    b. Assimilation efficiency
    c. Production efficiency
    d. All efficiency steps are equally affected

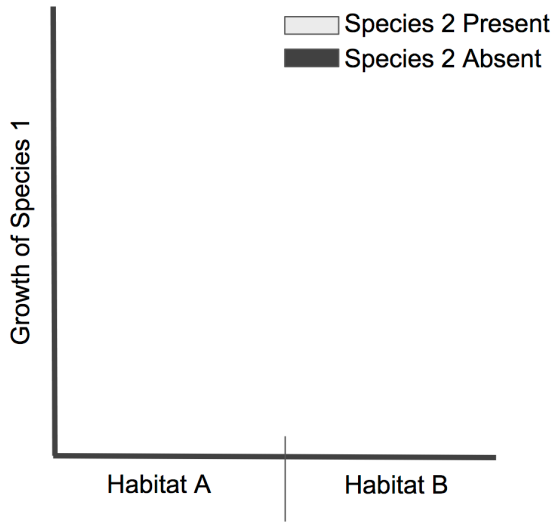
6. Explain why positive interactions are more likely or common under "stressful" conditions. Draw a graph depicting how the strength of positive interactions changes along some environmental stress gradient (predation pressure, physical stress, etc).

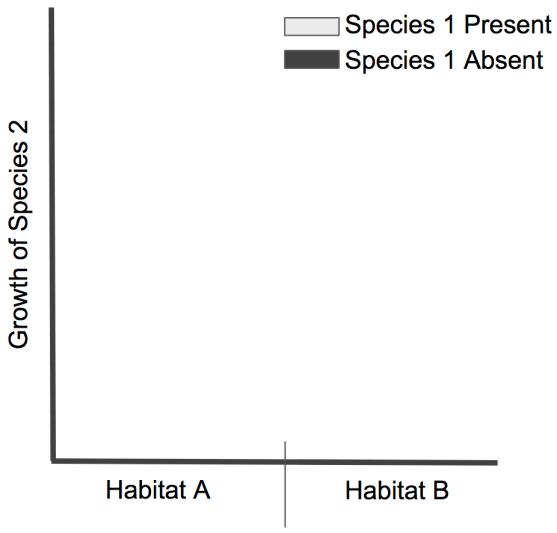
7. Using a specific example of an interspecific interaction, describe how the same two species can interact to produce a range of outcomes, from positive (facilitative, mutualistic, etc) to negative (competitive, parasitic, etc). What components of the physical or biotic environment determine whether the interaction is positive or negative?

8. Describe the differences between primary and secondary succession and give an example of the type of disturbance that would lead to each pattern of succession.

9. What are the characteristics of typical early successional species and late successional species?

10. The graphs below show growth of Species 1 in Habitats A and B when Species 2 is present and absent (left graph) and the growth of Species 2 in Habitats A and B when Species 1 is present and absent (right graph). Complete the bar graphs with data that represent a conditional Interaction.



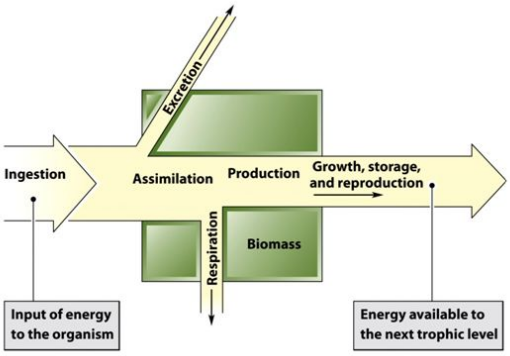


11. Ecosystem A has primary production of 1000 g C /m^2/yr and ecological efficiency of 10%. Ecosystem B has primary production of 300 g C /m^2/yr and an ecological efficiency of 25%.
    a. Which ecosystem will have more production at the secondary consumer (carnivore) trophic level?
    b. Which ecosystem is more likely to support an endothermic primary consumer (herbivore)?

c. If a trophic level requires at least 1 $gC/m^2/yr$ in order to exist, how many trophic levels can each of these ecosystems support?Given your answer to part c, which do you think has a larger impact on the energy available at upper trophic levels: primary productivity or efficiency of energy movement across trophic levels?

12. Describe how the following terms relate to one another: consumption efficiency, assimilation efficiency, production efficiency, and ecological efficiency.

13. The diagram at right describes how energy moves through a trophic level. Recall the definitions of the different types of efficiencies to answer the questions below



. Would you expect the assimilation efficiency to be higher for a carnivore or an herbivore?

a. Would you expect the production efficiency to be higher for an ectotherm or an endotherm?

b. Would you expect the consumption efficiency of primary consumers to be higher in a forest or a grassland?

c. Which would you expect to have higher production efficiency: a C3 plant or a C4 plant?

14. Why is terrestrial net primary productivity higher in the tropics, but marine net primary productivity is higher in coastal areas?

15. You may have heard it argued that being a vegetarian is more energy efficient than eating meat. Use your knowledge of energy transfer across trophic levels to describe whether you accept or reject this argument.

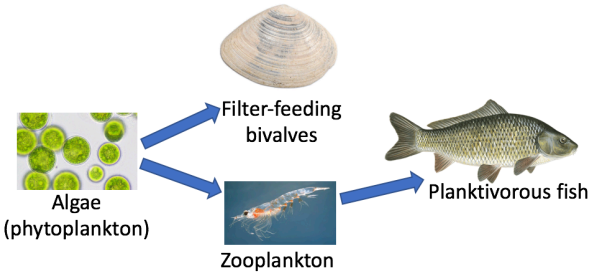
16. What are dead zones and why do they commonly form at the mouth of large rivers?

17. The diagram of the hydrologic cycle states that groundwater plays a small role in the global water cycle because it is 'locked in' to rock and soil layers. How is groundwater important for humans? How does the disconnect of groundwater from the global water cycle impact our future?

18. In class, Laci stated that diagrams of energy across trophic levels must always have a pyramid shape (most energy in the producers, decreasing at each level up to the top predators), but that diagrams of biomass or numbers of individuals in each trophic level can be a variety of shapes. Describe why energy diagrams must be a pyramid, and describe examples of ecosystems that may have biomass or number of individuals that are not pyramid-shaped.

19. You are an environmental consultant hired by a town adjacent to a small lake in the Sierras. The figure below describes what you know about the food web of the lake. The town is economically depressed, and residents have proposed two ideas for stimulating the local economy (1)

introduce northern Pike (a fish that only eats other fish) into the system to attract sport fishers (and their money) to the area and (2) harvest filter-feeding bivalves from the lake for food (they are quite a delicacy and fetch a hefty price). The town wants you to predict for them the consequences of each proposal for the health of their lake, long renowned for its clear waters.



20. What is the difference between a top-down trophic cascade and a bottom-up trophic cascade? Give an example of each.

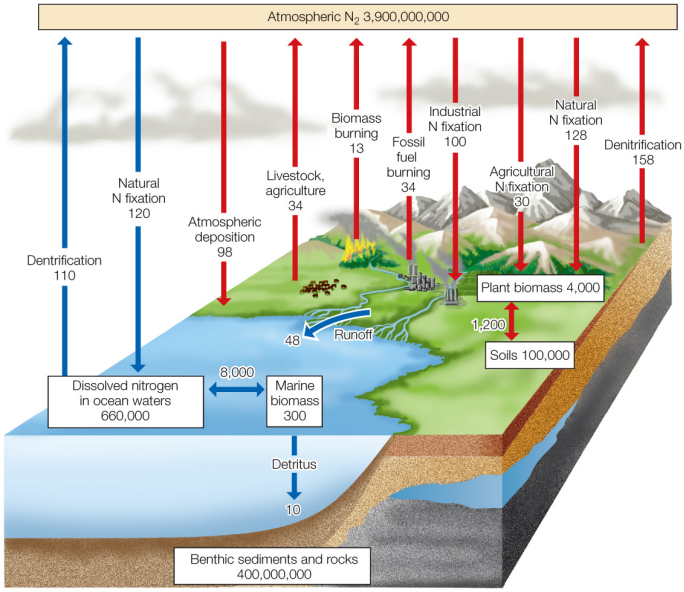
21. In class, Laci referred to the Ogallala Aquifer as 'fossil water.' How are aquifers similar to other 'fossil' resources like oil and coal?

22. Use the diagram below of the global nitrogen cycle to answer the following questions. *Note: for the nutrient cycle diagrams, I do **NOT** want you to memorize pools and fluxes, I **DO** want you to be able to read and interpret a cycle diagram*

    a. What is the largest flux? What is the largest pool?

    b. Which of the fluxes are anthropogenic (caused by humans)? How do these fluxes compare in magnitude to the natural fluxes?

    c. Describe a pathway by which a molecule of nitrogen in the atmosphere could move through both terrestrial and aquatic pools and then return to the atmosphere.



LIFE: THE SCIENCE OF BIOLOGY 11e, Figure 57.17
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